📘 Chapter 1 – Computer Networks

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1. How Networks Developed

What is a Network?

A **network** is a group of computers or devices that are connected to share data and resources.

How It Started (1980s)

- In the **1980s**, four universities were connected using **Ethernet cables**.
- These connections formed a **chain**, and that chain became the early form of a **computer network**.
- With time, more devices were added and the concept of a **global network → the Internet** began to evolve.

Why Networks Were Needed?

- To share information quickly.
- To reduce the use of physical storage (like floppy disks).
- To communicate across long distances.

Daily Life Example

- Using WhatsApp, Instagram, or Google on your phone → all depend on networks.

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2. Internet vs Intranet

Internet

- The **Internet** is a *public* network.
- It allows **global data sharing**.
- Anyone connected to the network can access public information.

Intranet

- The **Intranet** is a *private* network.
- Used inside companies or organizations.
- Only authorized people can access private data.

Daily Life Example

- **Internet**: Browsing YouTube.
- **Intranet**: Employees in a company accessing internal HR portal.

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3. Types of Networks

1. LAN (Local Area Network)

- Small area: home, office, school.
- Very fast and inexpensive.

Example: Wi-Fi in your house.

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2. MAN (Metropolitan Area Network)

- Covers a **city or large campus**.
- Boundary is *not fixed* (depends on the city/organization).

Example: Internet service provider (ISP) network inside a city.

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3. WAN (Wide Area Network)

- Covers **countries or continents**.
- Largest type of network.

Example: The global Internet is a WAN.

Difference: MAN vs WAN

Feature	MAN	WAN
Coverage	City level	Country or world level
Speed	Faster	Slower due to long distance
Ownership	Usually one organization/ISP	Shared by many companies
Boundaries	Not strictly fixed	Can span across nations

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4. IP Address

What is an IP Address?

- A unique **logical / virtual address** given to each device in a network.
- It identifies your device on the Internet.

Why is IP Address Used?

- To identify your device.
- To send/receive data to the correct destination (like a postal address).

How to Check IP/MAC Address

- **Get MAC Address:**
  getmac (in Command Prompt)
- **Get IP Address:**
  ipconfig /all

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5. MAC Address

What is MAC Address?

- A **physical** address burned into the network card.
- Unique for every device.
- Used by switches to deliver data.

Example: Like a unique fingerprint of your device.

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6. Network Devices

A. Router

What is it?

- A router **routes** (directs) data packets between networks.
- Connects private network to public Internet.

How it Works?

- Checks the **destination IP** of each packet.
- Sends it through the correct path like a GPS.

Why Used?

- To connect multiple networks.
- To provide Internet access.
- To manage traffic and security.

Daily Life Example

- Your home Wi-Fi router that connects your phone/laptop to the Internet.

B. Switch

What is it?

- An **intelligent device** used inside local networks (LAN).

How it Works?

- Creates a **MAC address table**.
- When a device sends data:
    - The switch learns **Sender MAC + Sender IP**.
    - When data comes back, the switch checks **Destination MAC**.
    - Sends data to the correct device **only**.

This process is called **MAC Learning**.

Why Used?

- To reduce network traffic.
- To improve security.
- Faster communication.

Daily Life Example

- Office computers connected to a switch for internal communication.

C. Hub

What is it?

- A **non-intelligent** device.
- Sends data to **all connected devices**, not just the correct one.

Why It's Problematic?

- Causes **privacy issues**.
- Creates **unnecessary traffic**.
- Very slow.

Daily Life Example

- Like a delivery boy going to **every house** in a society asking "Is this your parcel?"
  → Privacy is breached.

Current Use

- Hardly used today; replaced by **switches**.

D. Firewall

What is it?

- A **security device/software** that filters network traffic.
- Decides which data packets are allowed or blocked.

How it Works?

- Uses rules to check:
    - Source IP
    - Destination IP
    - Port number
    - Protocol

Why Used?

- To protect against hackers, malware, and attacks.
- To allow only safe connections.

Daily Life Example

- Blocking harmful websites on school or office networks.

🧠 1️⃣ CHAPTER SUMMARY

🌐 What is a Computer Network?

A **computer network** is a collection of computers and devices connected together to **share data, resources, and services** efficiently.

📌 Example: Mobile phones using Wi-Fi, offices sharing printers, using Google or WhatsApp.

🕰️ How Networks Developed (1980s)

- In the **1980s, four universities** were connected using **Ethernet cables**.
- These connections formed a **chain-like structure**.
- Gradually, more computers were added.
- This evolution led to the creation of the **Internet (global network)** 🌍

❓ Why Networks Were Needed?

✓ Faster data sharing
✓ Reduced physical storage (floppy disks, CDs)
✓ Long-distance communication
✓ Cost and time efficiency

🌍 Internet vs 🏢 Intranet

Feature	Internet 🌐	Intranet 🏢
Access	Public	Private
Scope	Global	Organization
Security	Less controlled	Highly secure
Example	YouTube, Google	Company HR portal

🧩 Types of Networks

🔹 **LAN (Local Area Network)**

- Small area (home, office, school)
- Very fast & low cost
- Example: Home Wi-Fi 🏠 📶

🔹 **MAN (Metropolitan Area Network)**

- Covers a **city or campus**
- Managed by ISP or organization
- Example: City ISP network 🌆

🔹 **WAN (Wide Area Network)**

- Covers **countries or continents**
- Slower due to long distance
- Example: The Internet 🌍

🔍 Difference: MAN vs WAN

Feature	MAN	WAN
Coverage	City	Country/World
Speed	Faster	Slower
Ownership	Single org/ISP	Multiple orgs
Boundary	Flexible	International

🧾 IP Address

- A **logical/virtual address**
- Identifies a device on a network
- Required for **sending & receiving data**

📌 Example: Like a **postal address** 📬

Command to check IP:
ipconfig /all

🧬 MAC Address

- **Physical address**
- Burned into network card
- Unique for every device
- Used by switches

📌 Example: **Device fingerprint** 👆

Command:
getmac

🛠️ Network Devices

📡 Router

- Connects private network to Internet
- Routes packets using IP address
- Acts like **GPS for data** 🧭

🔀 Switch

- Intelligent LAN device
- Uses **MAC Learning**
- Sends data only to correct device
- Faster & more secure ⚡ 🔐

🔁 **Hub**

- Non-intelligent device
- Broadcasts data to all
- Causes traffic & privacy issues ❌
- Almost obsolete

🔥 **Firewall**

- Security device/software
- Filters traffic using rules
- Protects against hackers & malware 🛡️

🎯 2️⃣ CONCLUSION

Computer Networks are the **backbone of modern digital communication** 🌐.
From small home networks (LAN) to the global Internet (WAN), networks enable **fast, secure, and reliable data exchange**.

Understanding **IP, MAC addresses, routers, switches, hubs, and firewalls** is essential for:
✓ Cyber Security
✓ Networking Jobs
✓ System Administration
✓ Interviews & Certifications

👉 **Without networks, today's digital world would not exist.**

🧩 3️⃣ MIND MAP

```
Computer Networks
|
├── Definition
|   └── Connected devices sharing data
|
├── History
|   └── 1980s → Universities → Internet
|
├── Need
|   ├── Fast sharing
|   ├── Less storage
|   └── Long-distance communication
|
├── Internet vs Intranet
|   ├── Public vs Private
|   └── Global vs Organization
|
├── Types
|   ├── LAN → Home/Office
|   ├── MAN → City
|   └── WAN → World
|
├── Addresses
|   ├── IP → Logical
```

```
|   └── MAC → Physical
|
└── Devices
    ├── Router → Route packets
    ├── Switch → MAC learning
    ├── Hub → Broadcast
    └── Firewall → Security
```

💼 🗣️ Q & A

❓ **Q1. What is a computer network?**

✅ **Answer:**
A computer network is a group of interconnected devices that communicate and share data and resources.

📌 **Why interviewer asks:** To test basic networking understanding.

❓ **Q2. How did computer networks start?**

✅ **Answer:**
In the 1980s, four universities were connected using Ethernet cables. This small network evolved into the Internet.

❓ **Q3. Why are networks needed?**

✅ **Answer:**
Networks are needed for fast data sharing, long-distance communication, reduced storage use, and cost efficiency.

❓ Q4. Difference between Internet and Intranet?

✅ **Answer:**
Internet is public and global, while Intranet is private and used within organizations.

❓ Q5. What is LAN, MAN, and WAN?

✅ **Answer:**

- LAN: Small area (home, office)
- MAN: City level
- WAN: Country or global level

❓ Q6. Which network is fastest?

✅ **Answer:**
LAN is the fastest because it covers a small area and has less delay.

❓ Q7. What is an IP address?

✅ **Answer:**
An IP address is a logical address used to uniquely identify a device on a network.

❓ Q8. Why is IP address important?

✅ **Answer:**
It helps in identifying devices and delivering data to the correct destination.

❓ Q9. What is a MAC address?

✅ **Answer:**
A MAC address is a physical address permanently assigned to a network interface card.

❓ **Q10. Difference between IP and MAC address?**

✅ **Answer:**

IP Address	MAC Address
Logical	Physical
Can change	Fixed
Used on Internet	Used in LAN

❓ **Q11. What is a router?**

✅ **Answer:**
A router connects different networks and forwards data packets using IP addresses.

❓ **Q12. What is a switch?**

✅ **Answer:**
A switch is an intelligent LAN device that sends data only to the intended device using MAC learning.

❓ **Q13. What is MAC learning?**

✅ **Answer:**
MAC learning is the process by which a switch stores MAC addresses to forward data efficiently.

❓ **Q14. Difference between hub and switch?**

✅ **Answer:**

Hub	Switch
Broadcasts data	Sends selectively
Non-intelligent	Intelligent
Slow	Fast
Less secure	More secure

❓ **Q15. Why hubs are not used today?**

✅ **Answer:**
Because they cause traffic congestion, privacy issues, and are very slow.

❓ **Q16. What is a firewall?**

✅ **Answer:**
A firewall is a security system that monitors and filters incoming and outgoing network traffic.

❓ **Q17. How does a firewall work?**

✅ **Answer:**
It checks IP address, port number, and protocol based on predefined rules.

❓ **Q18. Can firewall be hardware and software?**

✅ **Answer:**
Yes, firewalls can be hardware-based or software-based.

❓ **Q19. Which device provides Internet at home?**

✅ **Answer:**
A router provides Internet connectivity at home.

❓ **Q20. Why is networking important in cyber security?**

✅ **Answer:**
Because understanding networks helps in detecting attacks, securing data, and preventing unauthorized access.